

Task 3:

Bug 2

```
def get_monthly_attendance(conn, employee_id, month, year):
    query = """
        SELECT a.date, a.check_in, a.check_out, e.name
        FROM attendance a
        INNER JOIN employees e
            ON a.employee_id = e.id
        WHERE a.month = ?
            AND a.year = ?
            AND a.employee_id = ?
        ORDER BY a.date ASC
    """

    cursor = conn.cursor()
    cursor.execute(query, (month, year, employee_id))
    return cursor.fetchall()
```

Bug 3

```
from datetime import datetime

WORK_START = "09:00"
LATE_THRESHOLD_MINUTES = 15

def calculate_hours(check_in: str, check_out: str):
    fmt = "%H:%M"
    ci = datetime.strptime(check_in, fmt)
    co = datetime.strptime(check_out, fmt)

    duration = co - ci
    hours_worked = duration.seconds / 3600

    start = datetime.strptime(WORK_START, fmt)
    late_by = (ci - start).seconds // 60

    is_late = late_by >= LATE_THRESHOLD_MINUTES

    return round(hours_worked, 2), is_late
```

Bug 4

```
def generate_summary(records):
    total_hours = 0
    late_days = 0
    for i in range(len(records)):
        record = records[i]
        total_hours += record["hours"]
        if record["is_late"]:
            late_days += 1
    if len(records) == 0:
        avg_hours = 0
    else:
        avg_hours = total_hours / len(records)

    return {
        "total_hours": round(total_hours, 2),
        "avg_hours": round(avg_hours, 2),
        "late_days": late_days,
        "days_present": len(records)
    }
```

Bug 5

```
MIN_DAYS_REQUIRED = 20
MAX_LATE_DAYS = 3

def check_attendance_policy(summary):
    days_present = summary["days_present"]
    late_days = summary["late_days"]
    below_minimum = days_present < MIN_DAYS_REQUIRED
    exceeded_late = late_days >= MAX_LATE_DAYS
    if below_minimum or exceeded_late:
        return {
            "warning": True,
            "reason": [
                + (["Below minimum attendance"] if below_minimum else [])
                + (["Exceeded late check-ins"] if exceeded_late else [])
            ]
        }
    return {"warning": False, "reason": []}
```